A uniform C1 certificate

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The result below treats the complete state polynomial at a reciprocal measurement corner. Two nonnegative source weights control its state couplings through their sum, so either weight may vanish. The proof allows arbitrary relative size of h and σ while $1/(x\zeta)$ is bounded. We first give the finite tensor sum-of-squares construction used to complete the argument.

1 Exact sums of squares in the separated tensor ring

1.1 Two labelled fields and their residues

Let K_c, K_t be real closed extensions of $\mathbb{R}$ and put

$$T = K_c \otimes_{\mathbb{R}} K_t, \qquad B_i = \{x \in K_i : |x| < M \text{ for some } M \in \mathbb{R}_{>0}\}, \quad i \in \{c, t\}.$$

Write $\mathfrak{m}_i$ for the elements whose absolute value is smaller than every positive real number. The order on K_i is its unique field order. No order, embedding into a field, or algebraic-disjointness assumption is imposed on T.

Every $x \in B_i$ has a unique real standard part $\bar{x}$ with $x - \bar{x} \in \mathfrak{m}_i$. Indeed, the real cut $\{r \in \mathbb{R} : r \le x\}$ is nonempty and bounded; its supremum has the claimed property. Two such real numbers differ by a real infinitesimal and hence agree. Infinitesimals form an ideal of B_i, so $x \mapsto \bar{x}$ is an $\mathbb{R}$-algebra homomorphism with kernel $\mathfrak{m}_i$. In particular $B_i = \mathbb{R} \oplus \mathfrak{m}_i$ as real vector spaces.

Since tensoring real vector spaces preserves injections, $B := B_c \otimes_{\mathbb{R}} B_t$ embeds into T. Its residue map is

$$\rho : B \longrightarrow \mathbb{R}, \qquad \rho\Big(\sum_i a_i \otimes b_i\Big) = \sum_i \bar{a}_i \bar{b}_i. \tag{TF1}$$

The direct-sum decompositions above show that

$$\ker \rho = \mathfrak{m}_c \otimes B_t + B_c \otimes \mathfrak{m}_t. \tag{TF2}$$

Thus every tensor of zero residue has a finite expression in which each elementary term has at least one infinitesimal factor. None of these observations requires the two field extensions to be disjoint.

1.2 A strict real reserve gives an exact tensor sum of squares

Lemma TF1 (positive residue). If $a \in B$ and $\rho(a) > 0$, then a is a finite sum of squares in T. In fact its square roots may all be chosen to be elementary tensors.

Proof. By (TF2), write

$$a = \varepsilon + \sum_{j=1}^{N} x_j \otimes y_j, \qquad \varepsilon = \rho(a) > 0, \tag{TF3}$$

with bounded pure factors and an infinitesimal in at least one factor of each term. If $N = 0$, the assertion is immediate. Otherwise choose positive real numbers ε_j summing to ε. For a term with x_j infinitesimal, choose a positive real $M_j > |y_j|$. Direct multiplication gives

$$\varepsilon_j + x_j \otimes y_j = \frac{1}{2}\left[(\varepsilon_j/M_j + x_j) \otimes (M_j + y_j) + (\varepsilon_j/M_j - x_j) \otimes (M_j - y_j)\right]. \tag{TF4}$$

All four pure factors are strictly positive in their respective fields. Each elementary summand therefore is a square: for example $\frac{1}{2}u \otimes v = (\sqrt{u/2} \otimes \sqrt{v})^2$. If instead y_j is infinitesimal, interchange the roles of the two factors in this calculation. Sum (TF4) over j. Every operation is a field operation in one labelled factor; no mixed radical or mixed inverse occurs. □

Lemma TF2 (strict bounded Gram). Let $Q \in \mathrm{Sym}_r(B)$ and suppose its entrywise real residue $\rho(Q)$ is positive definite. Then

$$g^{\mathsf{T}} Q g \in \Sigma T^2 \qquad \text{for every } g \in T^r. \tag{TF5}$$

The entries of g need not be bounded, independent, or polynomial.

Proof. Choose a real invertible matrix C with $C^{\mathsf{T}}\rho(Q)C = I$. Replacing Q by $C^{\mathsf{T}}QC$ and g by $C^{-1}g$ preserves the quadratic expression, so assume $\rho(Q) = I$. Choose a real $\delta > 0$ such that $(r-1)\delta < 1$. With e_i the real coordinate vectors,

$$\begin{aligned} Q = &\sum_i \big(Q_{ii} - (r-1)\delta\big) e_i e_i^{\mathsf{T}} \\ &+ \sum_{i<j} \frac{\delta + Q_{ij}}{2} (e_i + e_j)(e_i + e_j)^{\mathsf{T}} \\ &+ \sum_{i<j} \frac{\delta - Q_{ij}}{2} (e_i - e_j)(e_i - e_j)^{\mathsf{T}}. \end{aligned} \tag{TF6}$$

The off-diagonal entries agree because the last two contributions have difference Q_{ij}; their diagonal contributions sum to δ for every incident pair. The residues of all scalar coefficients in (TF6) are positive. Lemma TF1 expresses each coefficient as a finite sum of squares in T. Evaluation on g then expresses (TF5) as a finite sum of squares of those scalar roots times g_i or $g_i \pm g_j$. This is a polynomial identity in the commutative ring T, so no size condition on g is involved. □

2 Large η with bounded reciprocal $x\zeta$: one proof for every h/σ

Let K_c, K_t be separately labelled real closed extensions of $\mathbb{R}$, $T = K_c \otimes_{\mathbb{R}} K_t$, and ΣT^2 the finite sums of squares in T. Bounded coefficients belong to $B_c \otimes B_t$ and their real residue is taken in both factors. We use the bounded-Gram fact: a bounded symmetric matrix with positive-definite real residue is a Gram matrix of a finite sum of squares in T. Roots in the Gram need not be bounded. A positive source or testing factor can be absorbed into a square using the real closedness of its own field.

Theorem 1. *Use the stationary C1 source with $d = 1$, $\lambda_{\mathrm{ev}} = n = 0$, $m, C, A, \alpha > 0$, $D, \beta \geq 0$, and the native strict gaps. Assume explicitly that*

$$x = k^2 \in \mathrm{Inf}_{c,>0}, \qquad \sigma = 1/\eta \in \mathrm{Inf}_{c,>0}, \qquad z = 1/(x\zeta) > 0 \text{ is bounded.}$$

There is no size restriction on $H = h/\sigma$. Then the complete reciprocal parent is in ΣT^2 for every independent pair $u, w \in \mathrm{Inf}_t$ and every real or complex state. There is no state-ratio or testing-ratio restriction. The literal faces $D = 0$ and $\beta = 0$ are included.

Equivalently, the theorem applies to the rational source below under $0 < b < 1$, $0 < r = bz < 1$, $0 < h < 1$, $E, M, J > 0$, $D, \beta \geq 0$, and the stated assumptions on x, σ, z.

The source and its uniform estimates. The rational identities and complete parent expansions below are algebraic for every positive z with the native gaps. Only their boundedness estimates and the theorem require z bounded. Put

$$r = bz, \quad Q = 1 - b, \quad 0 < b, r < 1, \quad \theta = 1 - h, \quad e = xh/\sigma = xH, \quad E = 1 - e, \quad K = 1 + x,$$

$$R_s = 1 + xr, \quad M = R_s - xz\theta > 0, \quad J = 1 + xh - \sigma(1 - r) > 0,$$

$$P = \sigma + K + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h\theta z.$$

Define the opening-cleared products

$$p = \frac{z(1+xh)E + (1-r)(1+xhz)}{2P},$$

$$l = \frac{(1+\sigma r)\{eR_s(1+\sigma z) + ME\}}{2P}, \qquad \kappa = \frac{K(1+\sigma r)(1+\sigma z)}{2MP}.$$

The event coefficients are

$$\begin{aligned} m = \kappa x^2 z H, \quad C = \frac{Kz(1+\sigma r)E}{2(1-r)P}, \quad A = p/r, \quad \alpha = l/(zQ), \\ D = \frac{K(1+\sigma z)(1+\sigma r) - JP}{2JP}, \qquad \beta = \frac{K(1+\sigma r)(1+xhz) - \theta P}{2\theta P}. \end{aligned} \tag{P1}$$

Only the original nonnegative D, β sources are considered. Set $a = Az = p/b$, $\ell = \alpha z = l/Q$, and

$$H_0 = 1 + zQ, \quad B_s = P + xz(J - \sigma),$$

$$\mathcal{H} = R_s H_0 + z\{(1-r)xh(2+xr) - e[2 + x(1+h+r\theta)] + Ke^2\}.$$

Direct multiplication gives the two exact source identities

$$2\theta P\beta = hB_s - \sigma H_0, \tag{BE3}$$

$$2P(JD + x\theta\beta) = \sigma^2\mathcal{H}. \tag{BE4}$$

They contain the entire scale estimate required below. Initially $0 < e < 1$ follows from $E > 0$, so $\mathcal{H}$ is bounded; also $P, J, M, B_s \sim 1$. Eliminating β between the identities gives

$$eB_s = xH_0 + \sigma\mathcal{H} - 2PJD/\sigma \le xH_0 + \sigma\mathcal{H}. \tag{1}$$

Thus $e = O(x + \sigma)$ is infinitesimal. Consequently

$$\mathcal{H} - H_0 \in \mathrm{Inf}_c, \quad D = O(\sigma^2), \quad \beta\theta = O(h + \sigma), \quad p - H_0/2,\ l - 1/2,\ \kappa - 1/2 \in \mathrm{Inf}_c, \qquad H_0 = 1 + zQ \ge 1.$$

The quantities p, l, κ are bounded and positive, and $m = \kappa xze$ is infinitesimal. The individual positive coefficients $C, a = p/b, \ell = l/Q$ may be unbounded; they will stay in whole squares.

The following estimates do not assume that H is bounded. Set

$$\nu = (2D - 2\beta\theta + h)/\sigma, \quad \lambda = (8Dh + 4\beta\theta^2 + 2h^2)/\sigma - 4bp\sigma z, \quad K_A = 4\kappa H - \lambda,$$

and define

$$\lambda_0 = \lambda - 2H, \quad d_G = 4D/\sigma^2, \quad \rho = \lambda x/\sigma, \quad L_0 = \lambda\sigma.$$

The source identities imply

$$\nu = 2D/\sigma + [H_0 - ez(J - \sigma)]/P, \tag{2}$$

$$\lambda = 2H + 2e\theta z(J - \sigma)/P - 2\theta H_0/P + 8Dh/\sigma - 4bp\sigma z. \tag{3}$$

Hence ν, d_G, λ_0 are bounded, $\lambda_0 + 2\theta H_0$ is infinitesimal, and $L_0 - 2h$ is infinitesimal. For $K_A = 4\kappa H - \lambda$ one also has

$$K_A - \{2\theta H_0 - 2h(1-r)\} \in \mathrm{Inf}_c\,. \tag{4}$$

Here is a finite polynomial derivation of the only possible pole in this assertion. Put

$$N_\kappa = K(1 + \sigma r)(1 + \sigma z) - MP = xF_x + \sigma F_\sigma,$$

where $F_x = (N_\kappa - N_\kappa|_{x=0})/x$ and $F_\sigma = N_\kappa|_{x=0}/\sigma$ are polynomials in the bounded source arguments. Direct substitution gives $F_\sigma = r - 1 + \sigma rz$. Therefore $H(2\kappa - 1) = (eF_x + hF_\sigma)/(MP)$ has residue $-h(1-r)$, which proves (4). In particular K_A is bounded.

For later use write $R_\gamma = R_0 - 8pu^2$, where

$$R_0 = -4\{2(C + D)(r - 1) + 2p + r - 1\}.$$

There is the uniform estimate

$$R_0 = O(x + \sigma + e), \qquad \lambda > 0 \implies R_0\sqrt{\lambda} \in \mathrm{Inf}_c\,. \tag{5}$$

To verify the first estimate without any source-pole division, temporarily treat e as an independent variable. The product $N = JPR_0$ is the polynomial

$$-4\{-JKz(1 + \sigma r)(1 - e) + (r - 1)K(1 + \sigma z)(1 + \sigma r) + J[z(1 + xh)(1 - e) + (1 - r)(1 + xhz)]\}.$$

It vanishes identically at $x = \sigma = e = 0$. Consecutive differences in x, σ, e express it as $xN_x + \sigma N_\sigma + eN_e$, with bounded polynomial quotients. For the second estimate use $\lambda = O(1 + H)$ from (3) and $e = O(x + \sigma)$ from (1). The bound

$$R_0^2\lambda = O((x + \sigma)^2(1 + H)), \qquad (x + \sigma)^2 H = xe + 2\sigma e + \sigma h$$

has an infinitesimal right side. This proves the square-root estimate for every possible H.

The same source formulas also give the identities used by the complementary testing charts. With $B = B_s$, put

$$d_0 = \sqrt{2PJD/(\sigma^2\mathcal{H})}, \qquad b_0 = \sqrt{2Px\theta\beta/(\sigma^2\mathcal{H})}.$$

The source identity gives

$$D = O(\sigma^2), \quad x\theta\beta = O(\sigma^2), \quad JD + x\theta\beta \asymp \sigma^2. \tag{BE5}$$

Then exactly

$$d_0, b_0 \geq 0, \quad d_0^2 + b_0^2 = 1, \quad D = \frac{\sigma^2\mathcal{H}}{2PJ}d_0^2, \quad \beta = \frac{\sigma^2\mathcal{H}}{2Px\theta}b_0^2. \tag{BE6}$$

$$H = \frac{H_0}{B} + \frac{\sigma}{x}\frac{\mathcal{H}}{B}b_0^2. \tag{BE7}$$

$$m = m_0 + m_1 b_0^2, \quad m_0 = \kappa x^2 z H_0/B \asymp x^2 z, \quad m_1 = \kappa x\sigma z\mathcal{H}/B \asymp x\sigma z. \tag{BE8}$$

$$\frac{C(1-r)}{z} \sim \tfrac{1}{2}, \qquad Ar \sim H_0/2, \qquad \alpha Q z \sim \tfrac{1}{2}. \tag{BE9}$$

No inverse of a sphere coordinate is used. Another useful uniform source identity is $p - l \in zB_c$: with e temporarily independent, $2P(p-l)$ is a polynomial in bounded x, σ, z, b, h, e, and at $z = 0$ its two source numerators are both one ($e + (1-e) = 1$ for the second). Its finite quotient by z is therefore polynomial and bounded. Restoring $e = xh/\sigma$ proves the assertion through both radial openings.

The complete state polynomial. Write $t = u^2$, $\xi = w^2$, $D_u = 1 + t$, $D_w = 1 + \xi$, $q = \sqrt{z}$, $g = h - Q$. The original reciprocal parent is

$$\begin{aligned} F_w = {} & 4m(W - wX + uY - uwZ)^2 + 4CD_w(uW - Y)^2 + 4AD_uD_wY^2 \\ & + (4D+2)(uwW + uX - wY - Z)^2 + (4\alpha+2)D_u(Y - wZ)^2 \\ & + (4\beta+2)D_w(X + uZ)^2 - 2D_uD_wZ^2. \end{aligned} \tag{P4}$$

Use the homogeneous roots

$$\begin{gathered} N_x = X + uhZ, \quad N_y = (Y - bwZ)/q, \quad \Gamma = \sigma Z - uwW, \quad R = wZ/q - uqW, \\ A_\theta = q(t-x)W, \quad P_\beta = uZ - (x/\sigma)wW, \quad X_e = X + ewW, \quad V = uw(t-x)W, \\ J_C = N_y + bR, \qquad J_\ell = N_y - QR. \end{gathered}$$

The following definitions collect every remaining coefficient:

$$\begin{aligned} \nu &= (2D - 2\beta\theta + h)/\sigma, \qquad \lambda = (8Dh + 4\beta\theta^2 + 2h^2)/\sigma - 4bp\sigma z, \\ \lambda_0 &= \lambda - 2H = 4D(1+h)/\sigma - 2\theta\nu - 4bp\sigma z, \\ K_A &= 4\kappa H - \lambda, \qquad B_x = 8\kappa e, \\ A_y &= 8(p-l) + 4z\{(1+2m)g + 2Dh + 2p\sigma\}, \\ R_x &= -4\{2Db + (1+2m)g - 2\beta\theta\}, \\ R_\gamma &= -4\{2(C+D)(r-1) + 2p + r - 1 + 2pt\}, \\ \chi &= 8(pb + lQ) - 4h + 4eh(2D+1) - 8Dh \\ &\quad + 4z\{4Dbh + 4pb\sigma + (1+2m)g^2 + 2\beta\theta^2\}, \\ k_1 &= 2(1+2m)g^2 + 8Dhb + 8pb\sigma + 4\beta\theta^2, \end{aligned} \tag{SP3}$$

$$\begin{aligned} E_{yy} &= 2z\{(1+2m)t + (1+2D)\xi\}, \\ E_{RR} &= z\{8bD + 8bp\sigma + 4b\sigma(2C+2D+1)(r-1) + k_1 t + 2b^2(2D+1)\xi\}, \\ E_{yR} &= 4z\{t[2Dh + 2p\sigma + (1+2m)g] + b(2D+1)\xi + \sigma(2C+1)(r-1) + 2D[1 + \sigma(r-1)] + 2p\sigma\}. \end{aligned}$$

Direct expansion of the original parent, using only the displayed source identities, gives the full four-state identity

$$\begin{aligned} F_w = {} & \mathcal{E}_+ + [2(1+\xi) + 4m\xi]N_x^2 + [4a + 2z(2C+1)]J_C^2 + 4\ell J_\ell^2 \\ & + E_{yy}N_y^2 + E_{RR}R^2 + E_{yR}N_yR + (4D/\sigma^2)\Gamma^2 \\ & - 4\nu N_x(u\Gamma) + R_\gamma(N_y + bR)(qw\Gamma) \\ & - 4uqw(2m + 2D + 1)N_xN_y + uqwR_xN_xR + qwB_xN_xA_\theta + uA_yN_yA_\theta \\ & + K_AA_\theta^2 + \chi uA_\theta R + \mathscr{S}, \\ \mathscr{S} = {} & \lambda\sigma P_\beta^2 + (\lambda x/\sigma)(t-x)(\xi - \sigma z)W^2, \end{aligned} \tag{SP6}$$

where all terms in the following remainder are retained tensor squares:

$$\begin{aligned}\mathcal{E}_+ &= 4\beta D_w N_x^2 + 2(2D+1)(uX_e)^2 + 4bpzt\Gamma^2 \\ &\quad + 4a(D_uD_w - 1)J_C^2 + 4\ell t J_\ell^2 + 4Cz\xi J_C^2.\end{aligned}$$

This identity includes all four state coordinates and both complete Gamma normal terms.

All E coefficients are infinitesimal. The state-normal cross coefficients outside $\mathscr{S}$, namely $\nu, R_\gamma, R_x, B_x, A_y, \chi$, are bounded; so are K_A, d_G. The possibly unbounded scalar coefficient λ is handled below. Indeed the only C poles outside positive squares occur through $C(r-1)$; β occurs there through the bounded product $\beta\theta$. Every non-Gamma cross coefficient has an explicit u or w factor and is infinitesimal. These conclusions use the uniform estimates just proved, not a bound on H.

Since $4a, 4\ell > 1$, clip these two whole positive radial squares at one and retain their source excesses. The resulting radial form is

$$J_C^2 + J_\ell^2 = (N_y + bR)^2 + (N_y - QR)^2,$$

whose real residue is positive definite because $b + Q = 1$. The N_x coefficient has residue two. The exact testing sign gauges $(u, Y, Z) \mapsto (-u, -Y, -Z)$ and $(w, X, Z) \mapsto (-w, -X, -Z)$ allow $u, w \geq 0$; literal zero values are allowed.

A single construction for $\lambda > 0$. In this paragraph $\rho > 0$. Direct elimination of β in (BE3) gives the exact identity

$$\begin{aligned}S := d_G + \rho &= 2\mathcal{H}/P + d_G(1 - J + 2hx) + 2eH_0/P \\ &\quad - 2e^2z(J - \sigma)/P - 4bpxz.\end{aligned} \tag{6}$$

Thus $S - 2H_0$ is infinitesimal. In particular S is a positive bounded source unit and d_G, ρ are both bounded. Also $q_0 := \lambda k$ is bounded, since $q_0^2 = L_0\rho$, and $\lambda x = \rho\sigma$ is infinitesimal.

Define the separated monomials, keeping the signs of W, Z,

$$B_1 = \sqrt{\lambda\sigma}\, uZ, \quad B_2 = \sqrt{\rho}\, uwW, \quad B_3 = \sqrt{\lambda}\, x\sqrt{z}W, \quad B_4 = \sqrt{\lambda x}\, wZ/\sqrt{z}.$$

Exactly

$$\mathscr{C} := \mathscr{S} + \lambda x R^2 = \sum_i B_i^2 - 4\lambda xuwWZ.$$

If $WZ > 0$, change the overall state sign so $W, Z > 0$, put $G = (B_1B_2B_3B_4)^{1/4}$ and $d_i = B_i - G$. Then

$$\mathscr{C} = \sum_i d_i^2 + \mathscr{C}_+,$$

$$\mathscr{C}_+ = 2G\{(\sqrt{B_1} - \sqrt{B_2})^2 + (\sqrt{B_3} - \sqrt{B_4})^2 + 2[(B_1B_2)^{1/4} - (B_3B_4)^{1/4}]^2\} \in \Sigma T^2.$$

All radicals separate into factors in K_c and K_t. If $WZ \leq 0$, put $d_i = B_i$ and $\mathscr{C}_+ = -4\lambda xuwWZ \in \Sigma T^2$. In both cases the differences $d_i - d_j$ equal $B_i - B_j$.

Put $T_\theta = k\sqrt{z}(u-k)W$ and $\Gamma_* = \sqrt{d_G}\Gamma$. The following exact conductors require no inverse of D or sphere coordinate:

$$\sqrt{\rho}\, u\Gamma = k(d_1 - d_2) - (u-k)(d_2 - d_3) - \sqrt{\rho\sigma}\, T_\theta, \tag{7}$$

$$\begin{aligned}\sqrt{\rho}\sqrt{z}w\Gamma &= \sqrt{\rho}\sigma zR - (\sqrt{z}w + z\sqrt{\sigma})(d_2 - d_3) \\ &\quad + \sqrt{\rho}(\sigma z/k + \sqrt{\sigma}\sqrt{z}w/k)T_\theta.\end{aligned} \tag{8}$$

Use $1 = (d_G + \rho)/S$ to write

$$u\Gamma = \frac{\sqrt{d_G}u}{S}\Gamma_* + \frac{\sqrt{\rho}}{S}(\sqrt{\rho}\, u\Gamma).$$

After (7), every coefficient on $(\Gamma_*, T_\theta, d_i)$ is infinitesimal. For the other complete normal use

$$\sqrt{z}wR_\gamma\Gamma = R_0\sqrt{z}w\Gamma - 8p\sqrt{z}wu(u\Gamma),$$

and apply the same identity $1 = (d_G + \rho)/S$ only to the first term. In (8) its only potentially singular T_θ coefficients reduce exactly to

$$\frac{R_0\rho\sigma z}{kS} = \frac{R_0 q_0 z}{S}, \qquad \frac{R_0\rho\sqrt{\sigma}\sqrt{z}w}{kS} = \frac{(R_0\sqrt{\lambda})\sqrt{\rho}\sqrt{z}w}{S}.$$

They are infinitesimal by (5); every other coefficient visibly is too. Thus both entire Gamma normal terms have infinitesimal coefficients on a single frame, including $D = 0$.

It remains to handle a possibly signed K_A. The same construction will work for every value of this source coefficient. Put

$$c = \max(0, 1 - K_A), \qquad a_0 = k/\sigma, \qquad J_A = \sqrt{z}(u-k)W, \quad F_A = u(Z - a_0 wW),$$

$$E_u = \sqrt{u}J_A, \quad E_k = \sqrt{k}J_A, \qquad U_* = \sqrt{q_0}E_u, \quad K_* = \sqrt{q_0}E_k.$$

One direct multiplication gives, for every state and all source sizes,

$$\mathscr{S} = L_0 F_A^2 + 2q_0 E_u^2 + q_0 E_k^2 + 2q_0\sqrt{u}E_u R. \tag{9}$$

Average this with $\mathscr{S} = \mathscr{C} - \lambda x R^2$:

$$\mathscr{S} = \tfrac{1}{2}L_0 F_A^2 + U_*^2 + \tfrac{1}{2}K_*^2 + \sqrt{q_0 u}\, U_* R + \tfrac{1}{2}\mathscr{C} - \tfrac{1}{2}\lambda x R^2.$$

The two cross/radial coefficients are infinitesimal. The term $L_0 F_A^2/2$ is retained as a positive square.

If $c > 0$, (4) implies that h is a positive unit: in fact its residue is at least $1/4$. The inequality obtained from (1),

$$a_0^2(hB_s - \sigma H_0) \le \mathcal{H},$$

then proves a_0 bounded. Also $L_0 \sim 2h$ is a positive unit, so λ is unbounded. The exact angular certificate proved below is

$$\Delta_A := \varepsilon_C \mathscr{C} + u^2 R^2 + \varepsilon_G \Gamma^2 - A_\theta^2 \in \Sigma T^2, \tag{10}$$

where

$$\varepsilon_C = 2/\lambda + u + k, \qquad \varepsilon_G = 2a_0 u + (u+k)/L_0.$$

Both coefficients are infinitesimal whenever $c > 0$. Subtract $c(d_G/S)\Delta_A$ from the parent, keeping that nonnegative quantity separately. The angular part of the remainder is

$$(K_A + c)A_\theta^2 - c(\rho/S)A_\theta^2.$$

The first coefficient is a positive unit. The second square is an infinitesimal Gram correction on (U_*, K_*), by the exact relation

$$\rho A_\theta^2 = (\sqrt{a_0 u}\, U_* + \sqrt{a_0 k}\, K_*)^2. \tag{11}$$

The other subtractions have coefficients $cd_G\varepsilon_C/S$ on $\mathscr{C}$, $cd_G u^2/S$ on R^2, and $c\varepsilon_G/S$ on Γ_*^2; all are infinitesimal. If $c = 0$, none of these terms is used. This verifies the same construction in both situations without a new choice of roots or a source/test rate comparison.

Finally split the positive angular coefficient between A_θ and T_θ, retaining the exact nonnegative difference

$$A_\theta^2 - T_\theta^2 = z(u-k)^2 W^2(u^2 + 2uk) \in \Sigma T^2.$$

The complete bounded Gram on

$$(N_x, N_y, R, \Gamma_*, A_\theta, T_\theta, d_1, \ldots, d_4, U_*, K_*)$$

has positive-definite residue: its radial block is $J_C^2 + J_\ell^2$, the N_x pivot is two, the Gamma pivot is one, the angular pivots are positive, and the circuit and last two pivots are respectively $1/2, \ldots, 1/2, 1, 1/2$. All other entries have zero residue. The remaining circuit coefficient $1/2 - cd_G\varepsilon_C/S$ has positive real residue $1/2$. The one-dimensional bounded-Gram fact makes it a tensor SOS, so its product with the retained $\mathscr{C}_+$ is also a tensor SOS. Restore the retained positive squares and apply the bounded-Gram fact.

Proof of the angular certificate, including state signs. For $U, V, Z_* > 0$ and positive separated a_* set

$$\Psi = 1 + U^2V^2 + U^2Z_*^2 + V^2Z_*^2 - 4UVZ_*, \quad T_* = U^2 - 1,$$

$$Q_* = VZ_* - U, \quad R_* = Z_* - UV, \quad G_* = U(V - Z_*).$$

The polynomial identity

$$\begin{aligned}
&[2 + a_*(U+1)]\Psi + U^2Q_*^2 + [2U + (U+1)/a_*]R_*^2 - T_*^2 \\
&= (T_* + UQ_*)^2 + 2Q_*^2 + (U+1)(\sqrt{a_*}G_* - R_*/\sqrt{a_*})^2 \\
&\quad + a_*(U+1)(1 - VZ_*)^2 + 2a_*(U+1)VZ_*(U-1)^2
\end{aligned}$$

is verified by multiplication. Expand each $U+1$ factor before taking square roots. The right side is a finite separated SOS. For $W \neq 0$ substitute

$$U = u/k, \quad V = w/(\sqrt{\sigma}\sqrt{z}), \quad Z_* = \frac{\sqrt{\sigma}}{k\sqrt{z}}\frac{Z}{W}, \quad a_* = \lambda k, \quad M_* = k^2\sqrt{z}W.$$

Then $\mathscr{C} = \lambda M_*^2\Psi$, $A_\theta = M_*T_*$, $R = k\sqrt{z}WQ_*$ and $\Gamma = k\sqrt{\sigma}\sqrt{z}WR_*$. Multiplication by M_*^2 gives (10). These are legitimate separated factors even when an intermediate ratio is unbounded. If $Z_* < 0$, apply the displayed identity first to $|Z_*|$. Replacing $|Z_*|$ by Z_* increases Ψ by $8UV|Z_*|$ and each of Q_*^2, R_*^2 by $4UV|Z_*|$. The three resulting additional terms have positive separated coefficients and are SOS. Thus the same certificate holds for either state sign. At $Z_* = 0$ the original identity applies directly. If $W = 0$, $A_\theta = 0$, so (10) follows immediately from $\mathscr{C} \in \Sigma T^2$. No state coordinate is inverted there.

The case $\lambda \leq 0$. The definition of λ gives $H^2 \leq 2bpz$ and $\beta\theta^2/\sigma^2 \leq bpz$. Thus H is bounded, h is infinitesimal, $\theta \sim 1$, $M_\beta = \beta\theta/\sigma^2$ is bounded and $H - H_0, d_G - 2H_0$ are infinitesimal. In particular

$$L = \lambda/\sigma = 8(D/\sigma^2)h + 4M_\beta\theta + 2H^2 - 4bpz$$

is bounded. Direct multiplication gives

$$u^2\Gamma + V = \sigma u P_\beta,$$

$$A_\theta^2 + u\Gamma P_\beta + uA_\theta R = \sigma P_\beta^2 + (x/\sigma)(u^2 - x)(w^2 - \sigma z)W^2,$$

where $V = uw(u^2 - x)W$. Consequently

$$K_A A_\theta^2 + \mathscr{S} = 4\kappa H A_\theta^2 + Lu^2\Gamma^2 + L\Gamma V + \lambda u A_\theta R.$$

Add $1 + L^2$ times the exact zero

$$V^2 - \sigma u^2 A_\theta^2 - u\sigma(u^2 - x)A_\theta R + (u^2 - x)V\Gamma = 0.$$

The (Γ, V) real residue block is

$$\begin{pmatrix} 2\bar{H}_0 & \bar{L}/2 \\ \bar{L}/2 & 1 + \bar{L}^2 \end{pmatrix}, \qquad \det = 2\bar{H}_0 + (2\bar{H}_0 - 1/4)\bar{L}^2 > 0.$$

The N_x, radial and angular blocks are strict; every remaining cross coefficient, including both complete Gamma normal terms, is infinitesimal. The bounded-Gram fact proves this case, including $\lambda = 0$.

All constructions are literal on the testing axes. No inverse of D or β occurs, so both permitted source faces are included. The parent is real quadratic in the state; applying the proof to the real and imaginary parts proves the complex-state assertion. This completes the theorem.